

QUESTION DEMONSTRATION

JSS 1-3 | Applied Systems Garden

SET UP

Prepare hypothesis, variables, treatment/comparison, replicates and QC for stocks, flows, feedbacks and trade-offs. Assign sample/instrument IDs.

DEMONSTRATE

Demonstrate how to question according to protocol, including raw-data recording and one quality-control check.

ASK

What is the hypothesis? Which variable is the response? Why replicate? What is the effect? What uncertainty qualifies the conclusion?

TD-391 | OBSERVE -> CHECK -> CLOSE

OBSERVE

Check protocol fidelity, traceable raw data, units, replicate handling, analysis and scientific integrity.

CHECK LEARNING

Student quantifies the response, identifies variation/uncertainty and makes a qualified conclusion or management recommendation.

CLOSE

Review abnormal readings without deleting them, connect evidence to management, secure instruments and record follow-up.

LINK THE LEARNING

Record evidence in the GrowMeal journal -> assign the related Garden Mission Card -> use the matching practical/data/problem-solving assessment item.

HYPOTHESIZE DEMONSTRATION

JSS 1-3 | Applied Systems Garden

SET UP

Prepare hypothesis, variables, treatment/comparison, replicates and QC for stocks, flows, feedbacks and trade-offs. Assign sample/instrument IDs.

DEMONSTRATE

Demonstrate how to hypothesize according to protocol, including raw-data recording and one quality-control check.

ASK

What is the hypothesis? Which variable is the response? Why replicate? What is the effect? What uncertainty qualifies the conclusion?

TD-392 | OBSERVE -> CHECK -> CLOSE

OBSERVE

Check protocol fidelity, traceable raw data, units, replicate handling, analysis and scientific integrity.

CHECK LEARNING

Student quantifies the response, identifies variation/uncertainty and makes a qualified conclusion or management recommendation.

CLOSE

Review abnormal readings without deleting them, connect evidence to management, secure instruments and record follow-up.

LINK THE LEARNING

Record evidence in the GrowMeal journal -> assign the related Garden Mission Card -> use the matching practical/data/problem-solving assessment item.

TEST DEMONSTRATION

JSS 1-3 | Applied Systems Garden

SET UP

Prepare hypothesis, variables, treatment/comparison, replicates and QC for stocks, flows, feedbacks and trade-offs. Assign sample/instrument IDs.

DEMONSTRATE

Demonstrate how to test according to protocol, including raw-data recording and one quality-control check.

ASK

What is the hypothesis? Which variable is the response? Why replicate? What is the effect? What uncertainty qualifies the conclusion?

TD-393 | OBSERVE -> CHECK -> CLOSE

OBSERVE

Check protocol fidelity, traceable raw data, units, replicate handling, analysis and scientific integrity.

CHECK LEARNING

Student quantifies the response, identifies variation/uncertainty and makes a qualified conclusion or management recommendation.

CLOSE

Review abnormal readings without deleting them, connect evidence to management, secure instruments and record follow-up.

LINK THE LEARNING

Record evidence in the GrowMeal journal -> assign the related Garden Mission Card -> use the matching practical/data/problem-solving assessment item.

REPLICATE DEMONSTRATION

JSS 1-3 | Applied Systems Garden

SET UP

Prepare hypothesis, variables, treatment/comparison, replicates and QC for stocks, flows, feedbacks and trade-offs. Assign sample/instrument IDs.

DEMONSTRATE

Demonstrate how to replicate according to protocol, including raw-data recording and one quality-control check.

ASK

What is the hypothesis? Which variable is the response? Why replicate? What is the effect? What uncertainty qualifies the conclusion?

TD-394 | OBSERVE -> CHECK -> CLOSE

OBSERVE

Check protocol fidelity, traceable raw data, units, replicate handling, analysis and scientific integrity.

CHECK LEARNING

Student quantifies the response, identifies variation/uncertainty and makes a qualified conclusion or management recommendation.

CLOSE

Review abnormal readings without deleting them, connect evidence to management, secure instruments and record follow-up.

LINK THE LEARNING

Record evidence in the GrowMeal journal -> assign the related Garden Mission Card -> use the matching practical/data/problem-solving assessment item.

MEASURE DEMONSTRATION

JSS 1-3 | Applied Systems Garden

SET UP

Prepare hypothesis, variables, treatment/comparison, replicates and QC for stocks, flows, feedbacks and trade-offs. Assign sample/instrument IDs.

DEMONSTRATE

Demonstrate how to measure according to protocol, including raw-data recording and one quality-control check.

ASK

What is the hypothesis? Which variable is the response? Why replicate? What is the effect? What uncertainty qualifies the conclusion?

TD-395 | OBSERVE -> CHECK -> CLOSE

OBSERVE

Check protocol fidelity, traceable raw data, units, replicate handling, analysis and scientific integrity.

CHECK LEARNING

Student quantifies the response, identifies variation/uncertainty and makes a qualified conclusion or management recommendation.

CLOSE

Review abnormal readings without deleting them, connect evidence to management, secure instruments and record follow-up.

LINK THE LEARNING

Record evidence in the GrowMeal journal -> assign the related Garden Mission Card -> use the matching practical/data/problem-solving assessment item.

QUANTIFY DEMONSTRATION

JSS 1-3 | Applied Systems Garden

SET UP

Prepare hypothesis, variables, treatment/comparison, replicates and QC for stocks, flows, feedbacks and trade-offs. Assign sample/instrument IDs.

DEMONSTRATE

Demonstrate how to quantify according to protocol, including raw-data recording and one quality-control check.

ASK

What is the hypothesis? Which variable is the response? Why replicate? What is the effect? What uncertainty qualifies the conclusion?

TD-396 | OBSERVE -> CHECK -> CLOSE

OBSERVE

Check protocol fidelity, traceable raw data, units, replicate handling, analysis and scientific integrity.

CHECK LEARNING

Student quantifies the response, identifies variation/uncertainty and makes a qualified conclusion or management recommendation.

CLOSE

Review abnormal readings without deleting them, connect evidence to management, secure instruments and record follow-up.

LINK THE LEARNING

Record evidence in the GrowMeal journal -> assign the related Garden Mission Card -> use the matching practical/data/problem-solving assessment item.

ANALYZE DEMONSTRATION

JSS 1-3 | Applied Systems Garden

SET UP

Prepare hypothesis, variables, treatment/comparison, replicates and QC for stocks, flows, feedbacks and trade-offs. Assign sample/instrument IDs.

DEMONSTRATE

Demonstrate how to analyze according to protocol, including raw-data recording and one quality-control check.

ASK

What is the hypothesis? Which variable is the response? Why replicate? What is the effect? What uncertainty qualifies the conclusion?

TD-397 | OBSERVE -> CHECK -> CLOSE

OBSERVE

Check protocol fidelity, traceable raw data, units, replicate handling, analysis and scientific integrity.

CHECK LEARNING

Student quantifies the response, identifies variation/uncertainty and makes a qualified conclusion or management recommendation.

CLOSE

Review abnormal readings without deleting them, connect evidence to management, secure instruments and record follow-up.

LINK THE LEARNING

Record evidence in the GrowMeal journal -> assign the related Garden Mission Card -> use the matching practical/data/problem-solving assessment item.

DIAGNOSE DEMONSTRATION

JSS 1-3 | Applied Systems Garden

SET UP

Prepare hypothesis, variables, treatment/comparison, replicates and QC for stocks, flows, feedbacks and trade-offs. Assign sample/instrument IDs.

DEMONSTRATE

Demonstrate how to diagnose according to protocol, including raw-data recording and one quality-control check.

ASK

What is the hypothesis? Which variable is the response? Why replicate? What is the effect? What uncertainty qualifies the conclusion?

TD-398 | OBSERVE -> CHECK -> CLOSE

OBSERVE

Check protocol fidelity, traceable raw data, units, replicate handling, analysis and scientific integrity.

CHECK LEARNING

Student quantifies the response, identifies variation/uncertainty and makes a qualified conclusion or management recommendation.

CLOSE

Review abnormal readings without deleting them, connect evidence to management, secure instruments and record follow-up.

LINK THE LEARNING

Record evidence in the GrowMeal journal -> assign the related Garden Mission Card -> use the matching practical/data/problem-solving assessment item.

MANAGE DEMONSTRATION

JSS 1-3 | Applied Systems Garden

SET UP

Prepare hypothesis, variables, treatment/comparison, replicates and QC for stocks, flows, feedbacks and trade-offs. Assign sample/instrument IDs.

DEMONSTRATE

Demonstrate how to manage according to protocol, including raw-data recording and one quality-control check.

ASK

What is the hypothesis? Which variable is the response? Why replicate? What is the effect? What uncertainty qualifies the conclusion?

TD-399 | OBSERVE -> CHECK -> CLOSE

OBSERVE

Check protocol fidelity, traceable raw data, units, replicate handling, analysis and scientific integrity.

CHECK LEARNING

Student quantifies the response, identifies variation/uncertainty and makes a qualified conclusion or management recommendation.

CLOSE

Review abnormal readings without deleting them, connect evidence to management, secure instruments and record follow-up.

LINK THE LEARNING

Record evidence in the GrowMeal journal -> assign the related Garden Mission Card -> use the matching practical/data/problem-solving assessment item.

RECOMMEND DEMONSTRATION

JSS 1-3 | Applied Systems Garden

SET UP

Prepare hypothesis, variables, treatment/comparison, replicates and QC for stocks, flows, feedbacks and trade-offs. Assign sample/instrument IDs.

DEMONSTRATE

Demonstrate how to recommend according to protocol, including raw-data recording and one quality-control check.

ASK

What is the hypothesis? Which variable is the response? Why replicate? What is the effect? What uncertainty qualifies the conclusion?

TD-400 | OBSERVE -> CHECK -> CLOSE

OBSERVE

Check protocol fidelity, traceable raw data, units, replicate handling, analysis and scientific integrity.

CHECK LEARNING

Student quantifies the response, identifies variation/uncertainty and makes a qualified conclusion or management recommendation.

CLOSE

Review abnormal readings without deleting them, connect evidence to management, secure instruments and record follow-up.

LINK THE LEARNING

Record evidence in the GrowMeal journal -> assign the related Garden Mission Card -> use the matching practical/data/problem-solving assessment item.